

6 THE LEARNING CURVE CRASHING THE FOSSIL FUEL PARTY

If there is only one thing you take from this book, let it be this: never make predictions about energy without using *learning curves*. Learning curves explain more about our energy future than all the experts and policy debates combined. It changed my life and it will change yours.

But I hadn't found the learning curve yet when I first made the graph from the previous chapter. I just had that graph and a severe bout of imposter syndrome. Who the hell was I to claim I saw something that all these real experts seemed to miss?

6.1 How Learning Curves Changed My Life

I read some papers by academics like Harmon and Nemet on how solar was going down in price 20% for every doubling of production ([Harmon, 2000](#)) ([Nemet, 2006](#)). I immediately thought: this is just like Moore's law regarding computer chips that dominated my previous career! Solar cells are even made of the same materials as computer chips!

But academics can sometimes be a bit removed from the real world and I still felt that it was impossible that so few people seemed to notice this. The scientist Richard Swanson had been presenting this insight at conferences since 2003 ([NREL, 2004](#)), but this was long before *The Economist* coined it Swanson's law and popularized it in 2012. Long before Tony Seba wrote about *Clean Disruption of Energy and Transportation* ([Seba, 2014](#)).

I started to read everything I could find about learning curves for solar and wind until I finally found an industry paper on the topic. It was titled *Experience Curves for Energy Technology Policy* ([IEA, 2000](#)). Best of all: it was published by the IEA itself. The report convincingly proved that solar was the future I thought, and yet I had this graph proving that the IEA was denying it. How was that possible?

I suddenly thought back of one of my first experiences at work. All of a sudden my imposter syndrome evaporated and was replaced by a stubbornness that propels me to this day.

I remembered a meeting at KPN, the biggest Telecom company in the Netherlands. It was 1994 or so. I was allowed to present my ideas about this new phenomenon called *The Internet* to a group of hotshots. I wanted to show them why The Internet would become very important and eventually replace analog telephony. After my talk I looked around nervously and saw the venerable experts looking at me with pity. "We will set aside 5% of future capacity for these pesky modems of yours, because they often stay connected for long periods" they said to me. An olive branch to this foolish but enthusiastic youngster and his sponsor. Then I was immediately shooed out of the room as the adults continued to discuss important things. Fifteen years later my last project at KPN was called *All IP: We replaced the last bits of analog telephony with the Internet Protocol*. The last bits of the

stuff that these venerable experts thought was more important, swept up and put in the dumpster.

Now here was the IEA, exhibiting what felt to me like exactly the same kind of behaviour as those venerable telecom experts. I had seen this movie before! I wasn't going to let myself be shooed out of the room again. This time I would not go quietly into the night. I would not let the planet go to hell in a wastebasket. I would not allow other kids to have the same traumatic classroom experiences. Just because some experts were too stuck in the past, too arrogant and too self-important to admit they were wrong.

I got a paper published in the Dutch Financial Times in July 2007: "the promise of solar cells is that everyone will produce their own cheap energy", I said. The experts, Shell, the International Energy Agency, they all seem to be missing it, but if the trends hold solar electricity will cost 1 cent per kilowatthour in 2032. And now I'm writing this book. Twenty years after the thrill of being published in a newspaper. Fifty years after that unpleasant experience in the classroom. Hoping that *this time* people will listen.

What I've done in the past 20 years is study how we must use this historic chance to revamp the whole energy system to make optimal use of abundant solar energy. I've written Dutch books, started a large research program at the university and a small company that model how to do it cost effectively. A few years back I've realized that combining the learning in solar panels with the learning in batteries and power converters takes the solar revolution to the next level. That's why I needed to write this book. But I'm getting ahead of myself again.

First I want to show you how this learning curve works. I want to give you X-ray vision for the energy transition. By the end of this chapter, you will understand something that has eluded the world's foremost energy experts for decades. You will see why predictions keep failing, why solar and batteries keep winning, and why the future looks radically different from what most institutions expect.

6.2 Practice Makes Perfect (and Cheaper)

Imagine you are installing your first solar panel. The mounting brackets are unfamiliar. The wiring diagram might as well be hieroglyphics. You drop a connector, fumble with the tools, second-guess every step. Four hours later, you have installed a single panel.

Now imagine your tenth installation. You know which tools to grab first. Your hands remember the sequence. You have developed little tricks: this bracket goes easier if you angle it slightly, that wire clips in faster from the left side. Four hours later the entire roof is done.

Your hundredth installation? You barely think about it. Ninety minutes, and you are already planning lunch.

This is obvious, of course. Practice makes perfect. But something else is happening too, something less visible. Your tools got better because the manufacturer learned what installers actually need. The panels got easier to install because designers heard feedback from the field. The supply chain learned to stock the right parts in the right places. The whole ecosystem learned alongside you.

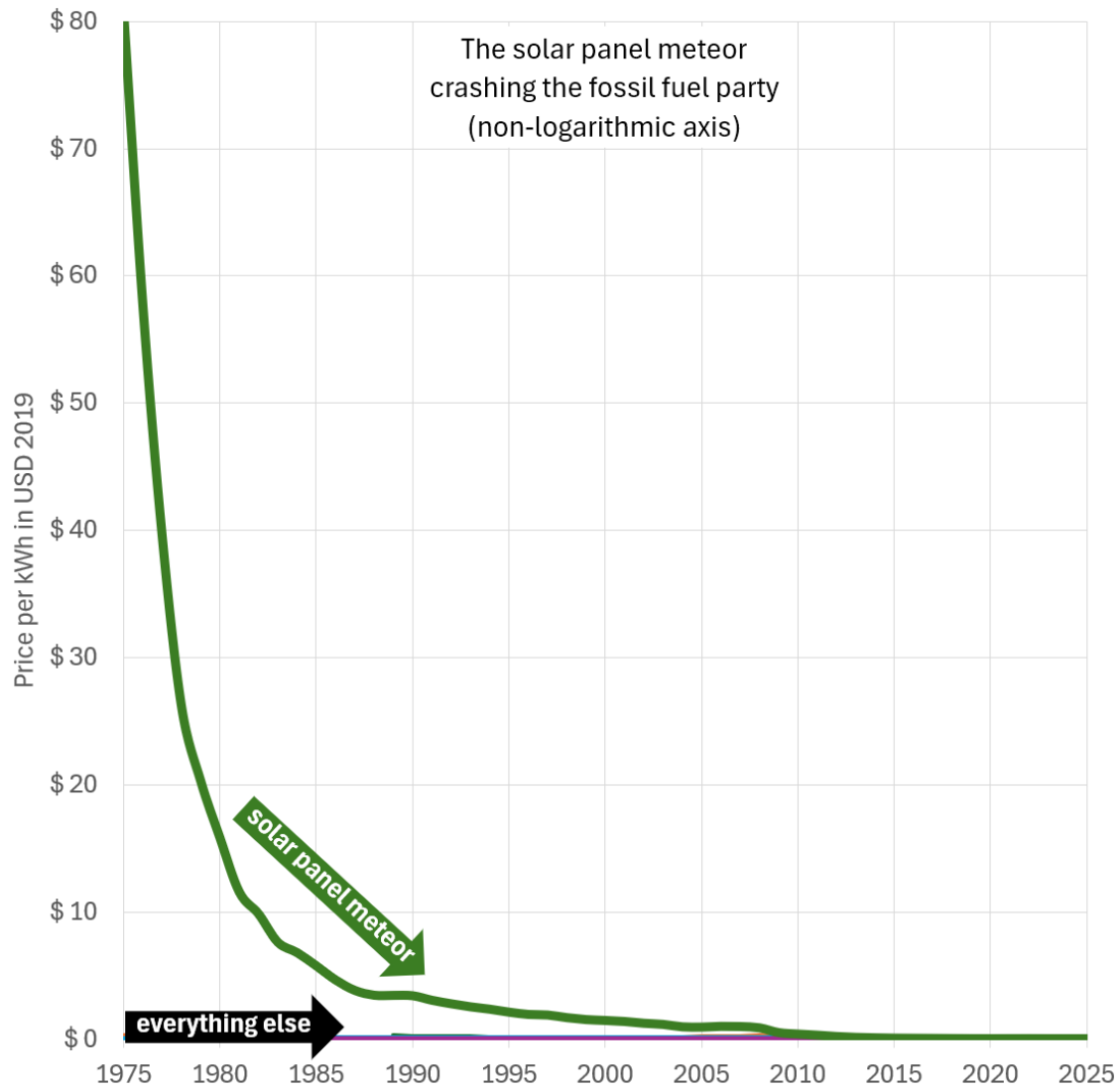
This happens at every scale: individual workers get faster, companies develop better procedures, industries accumulate shared knowledge, supply chains optimize themselves. Learning compounds like interest.

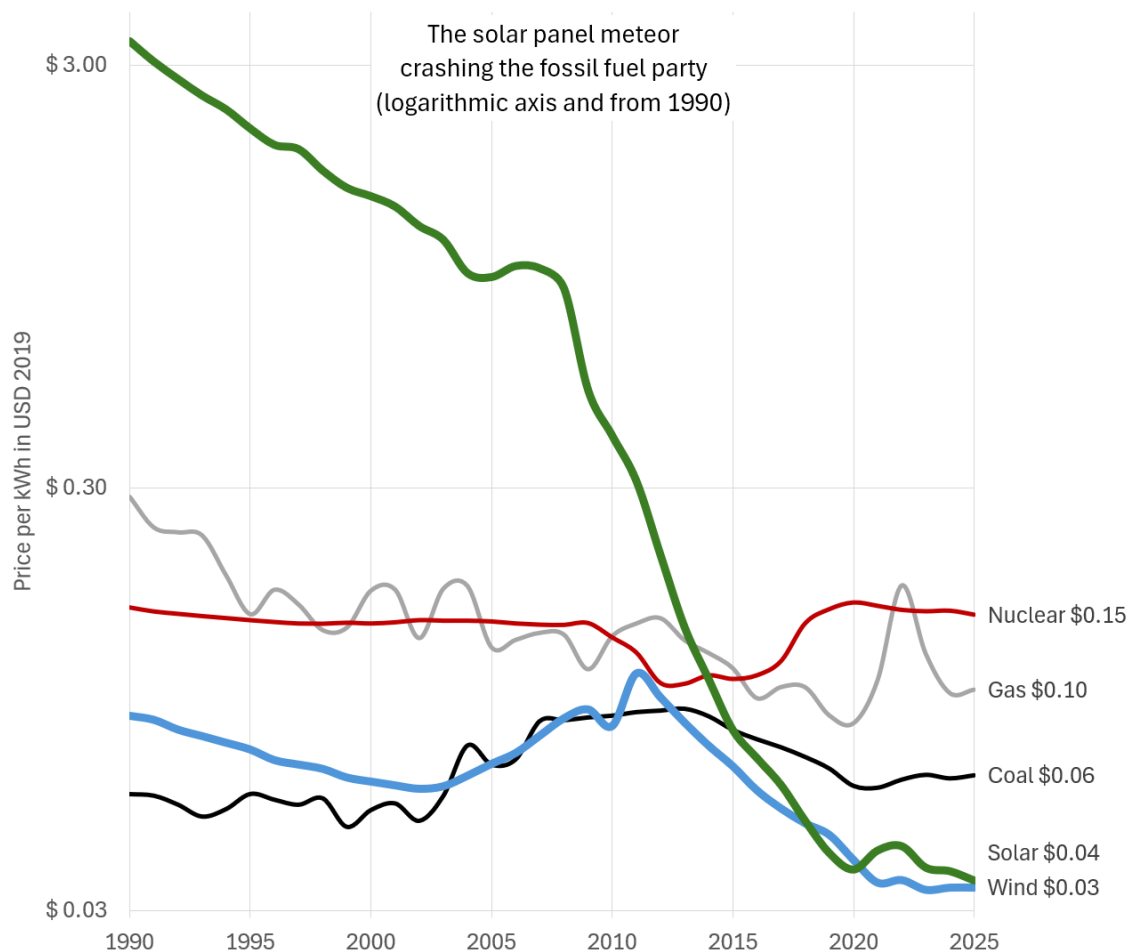
The remarkable thing is that this learning follows a *predictable pattern*. When humanity does something twice as many times, we get measurably, reliably better at it. Not by random amounts. By consistent percentages.

Most people know this intuitively. What most people do not know, including most energy experts, is what this pattern means for the future of energy.

6.3 The Solar Meteor Crashing the Fossil Fuel Party

Now that I had convinced myself that I could extrapolate the exponential growth, I could combine the learning curve from Wright's law to predict the price development per year in the future. Below is a figure that shows what *actually happened*. I call it "The solar panel meteor crashing the fossil fuel party." It's almost exactly the same as what the simple extrapolation showed my energy newbie self in 2007.





First you see the non logarithmic version. I like to think of it of showing the meteor crash clearly but from a distance. It started with solar being eye wateringly expensive in 1975. Back then it cost \$80 per single kilowatt hour of energy. On an average day my house and electric car need about 22 kilowatt hours. That means energy would cost me around \$2000 *a day* using these 1975 solar panels.

What happened next is hard to see on a non-logarithmic axis. That's why I've repeated the figure on the right. This time using an exponential axis and starting in 1990. It's like you are zooming in. You can see how the cost of solar kept plummeting. It followed Wrights learning curve almost perfectly as production scaled up. The result is that energy from solar panels now costs around \$0.04 per kilowatt hour ([IRENA, 2024](#)). Roughly \$1 per day for my house and electric car. That's 2000 times cheaper in 50 years.

Pause on that number. Imagine any other product in your life becoming 2000 times cheaper. A car going from \$50,000 in 1975 to \$25 in 2025. A house going from \$500,000 in 1975 to just \$250 in 2025. It's insane, right? No wonder the experts could not believe this was happening to the energy system they had known all their lives. But it was not insane. It was not random. It was not luck. It was not unpredictable. It was simply Wrights Law playing out decade after decade.

Only wind can keep up with solar somewhat, but even wind is going to be severely undercut in terms of costs by solar.

As you can see from the graph, fossil fuel isn't really learning. It's not keeping up ([Rubin, 2015](#)). That's no fault of the fossil fuel industry. It's an inherent and unavoidable result of the fact that fossil fuels are fully mature, don't scale up anymore, and need to burn lots of ever scarcer raw material.

I want to stress that this graph leaves out a lot. For starters, it leave out societal costs as quantified by empirical economists. Societal costs mean we put a number on people dying from particulate matter from burning fossil fuel, on footing the bill of climate change to next generations. Et cetera. In an ideally rational world this would double or quadruple the cost of fossil fuel ([Sovacool, 2021](#)) as we will see in part 3 of the book.

This graph also leaves out the benefits of solar for implementation. Everybody can simply put solar panels on their roof, even in places where there is no grid. Solar is not only the cheapest but also the simplest to implement, most distributed, most democratic, and hardest to control energy source in history.

In our capitalist society, many problems are attributable to capitalist benefits that do not match societal benefits. For solar, societal benefits and capitalist drive are aligned.

6.4 Wright's Law: The Pattern Behind Progress

In 1936, an engineer named Theodore Wright noticed something peculiar about airplane manufacturing ([Wright, 1936](#)). Every time the cumulative number of aircraft produced doubled, the cost per plane dropped by roughly 15%. Not randomly. Not sometimes. Every time.

This was not a policy. No one had decreed that planes should get cheaper at this rate. It emerged from the activity itself: from workers learning, from processes improving, from designs evolving through experience.

Wright had discovered a law of industrial learning. Every doubling of cumulative production yields a fixed percentage reduction in cost.

The pattern turns out to be remarkably robust. It applies to ships and semiconductors, to chemicals and cars. And it applies, with particular force, to the technologies of the energy transition.

Solar panels have followed Wright's Law with startling consistency. Since 1975, every doubling of cumulative solar production has reduced costs by approximately 20% ([Way, 2022](#)). In 1975, a watt of solar capacity cost about \$106. Today it costs around \$0.20 ([Nemet, 2006](#)). That is a 99.8% decline, tracking the learning curve almost exactly.

Wind turbines show a gentler but still powerful curve: roughly 12% cost reduction per doubling ([Wiser, 2016](#)). Wind started from a lower cost base than solar, so the absolute savings per doubling are smaller. But the pattern is the same. Modern wind turbines

produce electricity at a fraction of what they cost two decades ago, and the learning continues. Offshore wind, still early on its curve, is falling even faster as the industry masters the challenges of building in the ocean ([Tsiropoulos, 2018](#)).

Batteries show an even steeper curve: roughly 28% cost reduction per doubling ([Ziegler, 2021](#)). Lithium-ion batteries have fallen from over \$1,000 per kilowatt-hour in 2010 to under \$140 today ([BloombergNEF, 2019](#)). This is the technology that makes intermittent renewables dispatchable, that enables electric vehicles, that will eventually let your home ride through blackouts on stored sunshine. And it is still learning fast.

Digital power converters are the unsung heroes of the energy transition. Every solar installation needs an inverter to convert the panel's direct current into the alternating current your appliances expect. Every battery needs a bidirectional inverter to charge and discharge. Every electric vehicle charger, every grid-scale storage system, every wind turbine depends on power electronics to manage the flow of electricity.

These devices have followed the semiconductor learning curve, benefiting from decades of progress in digital electronics ([Farmer, 2016](#)). The same manufacturing improvements that made your smartphone cheap and powerful have made solar inverters cheap and reliable. A residential solar inverter that cost thousands of dollars fifteen years ago now costs a few hundred, and does more.

The beauty of this quartet is that they learn together. More solar and battery deployment means more inverter production, which means cheaper inverters, which makes solar and battery installations cheaper, which drives more deployment. Cheaper batteries make solar more valuable by storing midday surplus for evening use, which drives more solar deployment, which drives more battery deployment. The learning curves reinforce each other.

These are not projections or hopes. They are measurements, extracted from decades of data.

Twenty percent per doubling might sound modest. It is not. Doublings *compound*. Ten doublings in solar production means cost fall to 10% of where they started.

Wright's Law does not tell you when doublings will happen. That depends on demand, investment, and deployment. But it tells you what happens when they do. If you can estimate future production, you can estimate future costs with surprising accuracy.

This is the closest thing to a crystal ball that the energy world has. And for decades, the most influential energy institutions refused to use it.

6.5 Why the Experts Missed It

Why could outsiders like me see the pattern, while the International Energy Agency, with hundreds of analysts and decades of data, could not. This is not about intelligence. The experts I've met are smart people, often brilliant. So what went wrong?

Part of the answer lies in what Thomas Kuhn called *paradigm blindness*. ([Kuhn, 1962](#)) Experts trained in one paradigm struggle to see another emerging. The energy analysts who built their careers on fossil fuels understood extraction: finite resources, geological constraints, depletion curves. Their mental models assumed scarcity.

Learning curves assume something different. They assume improvement through practice. They assume that doing more of something makes you better at it. This is a fundamentally different way of thinking about energy, and it was almost invisible to people steeped in the old paradigm.

I had an advantage they did not: I came from a world where learning curves were the whole story. In telecom and computing, everyone understood that costs would fall with scale, that performance would improve with each generation, that today's expensive novelty would become tomorrow's commodity. This was not a revelation; it was the water we swam in. When I looked at solar, I saw something familiar. They looked at solar and saw something foreign.

There is also the problem of exponential growth itself. Humans intuitively think linearly: we underestimate exponential growth and perceive it in linear terms ([Wagenaar, 1975](#)) ([Schonger, 2021](#)) ([Teichman, 2022](#)). If something grew this much yesterday, our intuition expects it to grow about that much tomorrow. Exponential growth feels wrong, implausible, suspicious. Even people who understand exponentials intellectually struggle to feel them in their bones.

The IEA, again and again, assumed solar growth would slow down. Every year they bend down the exponential curve. This is what happens when human apply intuition to a domain where human intuition fails.

And then there is institutional inertia. Energy agencies advise governments. Governments plan infrastructure. Infrastructure takes decades. Admitting that your predictions were systematically wrong threatens your credibility. It is easier to adjust slightly than to admit the model is broken.

The cost of this collective blindness is difficult to calculate. By my count it's at least dozens of trillions in misallocated investment. Decades of delayed climate action. Countless decisions made on faulty projections. The world built gas plants and planned coal expansions when it should have been building solar farms and battery factories.

But recrimination is not the point. Understanding is. Now that you see what they could and maybe can not, you can evaluate energy forecasts for yourself. When someone tells you renewables are too expensive, or that growth will slow down, or that we will always need fossil fuels, you can ask: are they using learning curves? If not, why should you trust their numbers?

6.6 Why Solar Learns Faster: The Physics of Improvement

Wright's Law applies to many industries, but not all industries learn at the same rate. Solar's 20% learning rate is steep. Nuclear power, as we saw in the previous chapter, has a negative learning rate: newer plants cost more, not less ([Grubler, 2010](#)). What explains the difference?

The first answer lies in what engineers call material intensity: how much physical stuff you need per unit of useful output. Coal power requires enormous quantities of material. You have to mine the coal, transport it (often across continents), combust it, and deal with the ash and emissions. A coal plant needs massive boilers, extensive pollution controls, and constant fuel delivery. Every megawatt-hour requires hundreds of kilograms of material to be moved, processed, and disposed of ([EIA, 2023](#)).

The second answer lies in unit size: if it's a product that is produced again and again so there's plenty of opportunities to learn, or if it's a megaproject. Nuclear power plants are megaprojects and every time there is learning about what could go wrong, it potentially makes the next megaproject more expensive. It also requires extraordinary precision engineering: containment vessels, cooling systems, radiation shielding, multiple redundant safety systems. The materials may be smaller in volume, but the complexity is immense, and that complexity grows with every safety upgrade.

Solar is different. A solar panel is essentially a thin layer of processed sand, arranged to catch photons. That's it. No fuel to mine or transport. No combustion. No cooling towers. No emissions to scrub. The entire system is solid-state: no moving parts in the panel itself.

The result is startling: solar uses over 100 times less material per unit of energy delivered than fossil fuels ([Hertwich, 2015](#)). And that is before we start recycling, which is much harder to do with stuff you burn and blow into the atmosphere.

This matters because of where learning can happen. When raw materials dominate costs, learning is constrained. You can get better at mining coal, but digging is a pretty mature technology and with the amount of mass you need to move, coal is pretty close to its floor price. With megaprojects, most people will only do one project in their lifetime. There is very little potential for learning. But when material costs are low and research keeps bringing them down, when costs are dominated by fabrication and assembly that is repeated time and time again, learning is unleashed. Every step in the research and production process becomes a target for improvement.

Solar panels are, in this sense, closer to making microchips than to mining. They benefit from the same learning dynamics that gave us smartphones and laptops: the relentless refinement of semiconductor manufacturing ([Farmer, 2016](#)). Silicon, copper, aluminum, glass, and a bit of silver, combined with manufacturing techniques that improve year after year.

Digital power converters are very similar and resemble chip manufacturing even more closely. Recent breakthroughs in power transistors (like the tiny switches on your smartphone chip but larger and able to handle much more electricity) promise extremely powerful power converters with energy losses below 1% ([Microchip USA, 2025](#)). Low losses in turn means little heat which in turn means smaller devices using less material. Which in turn means they can become cheaper still.

This is why my telecom background let me see the pattern. I had watched semiconductor economics play out for a decade. Solar was the same story in a different costume.

Wind turbines sit somewhere in between. They have more mechanical complexity than solar panels: gearboxes, generators, blades that must withstand decades of stress. But they too benefit from manufacturing learning, from better materials, from designs refined through millions of operating hours. The 12% learning rate reflects this middle position: not as steep as solar, but still powerful over time.

Batteries surprised me. I had expected them to have lower learning curves because of the material costs. But it turns out this is the most vibrantly innovating field of all. Maybe because it's not about mechanical systems but about better chemistries and production processes where so much is still unknown. Expensive metals like cobalt and nickel are being replaced by iron and phosphate. The resulting batteries have material costs that are trending towards just \$10 per kWh ([Knehr, 2024](#)). And sodium batteries that replace lithium metal with sea salt have the potential to bring the material costs much lower still ([Stanford, 2017](#)).

This learning is not a policy outcome. Yes, governments investing in science stand at the basis of these breakthroughs. Yes, the German *energiewende* made it possible that solar panels scaled up when they were still expensive, and this drove their global price reduction ([Clean Coalition, 2024](#)). But no government decreed that solar should learn faster than coal. It emerges from physics, from the fundamental nature of what these technologies are and how they are made.

And this is why my optimism is not based on hope or faith. It is in the physics of the technology itself. The learning curves will continue because the physics allows them to continue. The only question is how fast we move down the curves.

6.7 Sidebar: The Math Behind Wright's Law

For those who want to verify the pattern, here is the mathematics. Wright's Law is expressed as:

$$C(n) = C(1) \times n^{-\alpha}$$

where $C(n)$ is the cost after n cumulative units of production, $C(1)$ is the cost of the first unit, and α is the learning coefficient.

The learning rate is related to α by the formula: learning rate = $1 - 2^{-\alpha}$. For solar panels, α is approximately 0.32, yielding a learning rate of about 20% (Way, 2022). This means costs fall to 80% of their previous level with each doubling of production.

You can test this yourself. Plot the logarithm of cost against the logarithm of cumulative production. If Wright's Law holds, you will see a straight line. The slope of that line is $-\alpha$. Researchers have done this for solar panels going back to 1975, and the line is remarkably straight (Way, 2022).

The math does not tell you when doublings will happen, only what happens when they do. Forecasting deployment is harder than forecasting learning. But once you know the learning rate, and once you have a deployment scenario, the cost trajectory follows.

This sidebar is for those who want to check my homework. The main text has everything you need.

6.8 Sidebar: When Growth Slows Down

A reasonable objection: nothing grows exponentially forever. At some point, solar growth must slow. When?

Technologies typically follow an S-curve: slow initial growth, then rapid expansion, then saturation as the market fills up. (Rogers, 1962) Refrigerators followed this pattern. So did televisions, mobile phones, and cars.

The question is not whether solar will saturate. It will, eventually, when most suitable surfaces have panels and most electricity comes from the sun. The question is where we are on the curve.

The answer, by almost any measure, is early. Solar provided less than 9% of global electricity in 2025 (Ember, 2025), and as the transition

progresses, more and more processes will be electrified. We can expect decades of growth before saturation.

The same applies to batteries and power converters. Three quarters of cars sold by the end of 2025 will still contain a combustion motor ([BloombergNEF, 2025](#)). Stationary storage is just beginning to scale. The learning has decades to run.

The experts kept predicting the slowdown would come next year. It kept not coming.

6.9 X-Ray Vision

You now have the framework that most energy experts have lacked: the understanding that clean energy costs follow predictable learning curves, that these curves emerge from physics rather than policy, and that institutions trained in the scarcity paradigm have systematically failed to see them.

This is your X-ray vision for the energy transition.

When someone tells you renewables are too expensive, you can ask: expensive compared to what, and at what point on the learning curve? When an institution predicts that clean energy growth will slow, you can ask: what is their model, and have they accounted for Wright's Law? When a pundit declares that we will always need fossil fuels, you can ask: are they extrapolating from today's costs or from the trajectory?

But frameworks are only useful if they match reality. So let me show you what happens when you apply this framework to actual history.

The next chapter presents a graph that captures two decades of forecasting failure. The gap between prediction and reality is the cost of paradigm blindness.

Now that you understand learning curves, you understand why the experts kept getting it wrong. So what does that mean for the rise of solar and wind?